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Solving Linear Equations

Gaussian elimination

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned} \quad \longrightarrow \quad \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Homogeneous: $Ax=0$

soln: $(x_1, x_2, \ldots, x_n) = (t_1, t_2, \ldots, t_n)$

e.g. $\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 + 2x_2 &= 3 \end{aligned} \longrightarrow$ soln: $\begin{aligned} x_1 &= 1 \\ x_2 &= 1 \end{aligned}$

Any system of linear equations has 0, 1, or infinitely many solutions.

$Ax = b$ $\quad A^{m\times n},\ x^{n\times 1},\ b^{m\times 1}$

Augmented matrix: $[A \mid b] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right]$

Ex. 1: $\begin{aligned} x_1 + 2x_2 + 3x_3 &= 3 \\ 2x_1 - x_2 + x_3 &= 1 \\ x_1 - x_2 + x_3 &= 0 \end{aligned}$

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 2 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{array}\right]$$

The main idea is to carry out algebraic operations that do not alter the solution set of the system of equations and produce successively increasingly simpler systems to a point where we can figure out if the system has solutions, how many, and what those are.

Elementary row operations

a. $R_i \leftrightarrow R_j$
b. $R_i \leftarrow \alpha R_i \quad \alpha \neq 0$
c. $R_i \leftarrow R_i + \alpha R_j \quad \alpha \neq 0$

e.g. $\begin{aligned} 2x_1 + 3x_2 + 3x_3 &= 6 \\ 2x_1 - x_2 + 2x_3 &= 0 \\ 2x_1 - x_2 - x_3 &= 0 \end{aligned}$

$$\Rightarrow [A \mid b] = \left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 2 & -1 & 2 & 3 \\ 2 & -1 & -1 & 0 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 2 & -1 & -1 & 0 \\ 2 & -1 & 2 & 3 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 2 & -1 & -1 & 0 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 0 & -4 & -2 & -6 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$R_3 \leftarrow \frac{1}{3} \cdot R_3$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 0 & -4 & -2 & -6 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 6 \\ -4x_2 - 2x_3 &= -6 \\ x_3 &= 1 \end{aligned}$$

$$\begin{aligned} 2x_2 + x_3 &= 3 \\ x_2 &= 1 \\ 2x_1 + 3(1) + (1) &= 6 \\ x_1 &= 1 \end{aligned}$$

soln: $(1, 1, 1)$

← So these have the same soln set as the original eqns.
Equivalent eqns.

Ex.2. $x_1 + x_2 + x_3 = 3$
$-x_1 - x_2 + x_3 = 1$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ -1 & -1 & 1 & 1 \end{array}\right]$$

$R_2 \leftarrow R_2 + 1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{array}\right]$$

$2x_3 = 4$
$x_3 = 2$

$x_1 + x_2 = 1$ eqn of line

$x_1 = 1 - x_2$

$x_2 = t \Rightarrow x_1 = 1 - t$

soln: $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1-t \\ t \\ 2 \end{pmatrix}$ ∞ly many sol^{n}s.

Reduced row echelon form (RREF)

$A^{m \times n}$ is in RREF:

i) Row with first non-zero # must be 1 → leading 1.
ii) Rows of entire zeros must be grouped together at the bottom.
iii) For 2 successive rows,

$$\begin{bmatrix} 1 & * & * & * \\ * & * & 1 & * \end{bmatrix} \leftarrow \text{leading 1 must be ahead of that of prev row.}$$

] Row echelon form (properties i, ii, iii hold) obtained using Gaussian elimination.

iv) Each col^n with leading 1 must contain 0's everywhere in the col^n.

Procedure: Gauss-Jordan elimination.

Ex. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$ r_1, r_2, r_3 all have leading 1's (no zero rows)

$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 2 & 0 & 5 \\ 0 & 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$ > zero rows

} RREF

$\left[\begin{array}{cccc|c} 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ ← REF

$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right]$ ← REF

Steps to solve Ax=b

1. Augment A with B. $[A|b]$
2. Row operations to get RREF.
3. Obtain soln set by:
   - inspection
   OR - parametric form.

Comments on RREF

$A^{m \times n}$

1) $Ax = 0$, # of leading 1's $< n$] $\Rightarrow$ $Ax=0$ has only many soln^s.

2) # of leading 1's $= n$] $\Rightarrow$ $Ax=0$ has unique (trivial) soln.

3) Every non-zero matrix has a unique RREF.
   For a non-zero matrix A,
   $\exists$ unique B s.t. A is row equivalent to B.

4) Given RREF(A), we can check consistency for $Ax=b$.

Ex. $[A|b] = \left[\begin{array}{ccccc|c} 1 & 0 & 2 & 1 & 0 & 5 \\ 0 & 1 & 3 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$

inconsistency $\Rightarrow$ NO soln.

Ex. $x_1 + x_2 + x_3 = 3$
$-x_1 - x_2 + x_3 = 1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ -1 & -1 & 1 & 1 \end{array}\right]$$

$R_2 \leftarrow R_2 + R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{array}\right]$$

$R_2 \leftarrow \frac{1}{2} R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 \leftarrow R_1 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$x_3 = 2$ (pivot or leading)

$x_1 + x_2 = 1$

$\Rightarrow x_1 = 1 - x_2$

$$\begin{pmatrix} 1-t \\ t \\ 2 \end{pmatrix}$$

x_1 and x_3 are PIVOT variables.
x_2 : FREE variable.

Rank of A

Rank(A) = # of pivot variables in RREF(A).

of pivots (rank) + # of free variables = # of colms (A) = n

Existence of free variables → only many sol^n's.

How do the elementary row operations work? (row operations as matrix multiplication)

i) $R_i \leftrightarrow R_j$ (row swap)

ii) $R_i \leftarrow kR_i$ $\quad k$: real scalar

iii) $R_i \leftarrow R_i + kR_j$

How do we carry these out?

Row swap:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$Ax = b$$

$[A|b]$ ← we perform row operations on augmented matrix.

- $$\left[\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} a_{21} & a_{22} & b_2 \\ a_{11} & a_{12} & b_1 \end{array}\right]$$

$$I_{2\times 2}\,[A|b] = [A|b]$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}_{2\times 2} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}_{2\times 1} \longrightarrow \begin{bmatrix} a_{21} \\ a_{11} \end{bmatrix}_{2\times 1}$$

Row swap:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \left[\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right] = \left[\begin{array}{cc|c} a_{21} & a_{22} & b_2 \\ a_{11} & a_{12} & b_1 \end{array}\right]$$

Replacing row with scalar multiple of same row:

$R_i \leftarrow kR_i$

$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ ka_{21} & ka_{22} \end{bmatrix}$$

$R_i \leftarrow R_i + kR_j$

$R_2 \leftarrow R_2 + kR_1$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21}+ka_{21} & a_{22}+ka_{12} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21}+ka_{11} & a_{22}+ka_{12} \end{bmatrix}$$

For $R_1 \leftarrow R_1 + kR_2$ $\quad \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$

Every elementary row operation is done using matrix multiplication.
- Pre-multiply with elementary matrices.

Why do the solutions stay the same?

$$a_{11}x_1 + a_{12}x_2 = b_1 \leftarrow l_1$$
$$a_{21}x_1 + a_{22}x_2 = b_2 \leftarrow l_2$$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$(A|b) = \left[\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right]$$

on row swap, $\left[\begin{array}{cc|c} a_{21} & a_{22} & b_2 \\ a_{11} & a_{12} & b_1 \end{array}\right] \begin{matrix} \leftarrow l_2 \\ \leftarrow l_1 \end{matrix}$ it's the same lines, so point of intersecⁿ stays same.

on scaling, eqⁿ still stays the same.

$R_i \leftarrow R_i + kR_j$:

$$2x_1 + x_2 = 3$$
$$x_1 + 2x_2 = 3$$

$$\left[\begin{array}{cc|c} 2 & 1 & 3 \\ 1 & 2 & 3 \end{array}\right]$$

$$R_2 \leftarrow R_2 + kR_1$$

$$\left[\begin{array}{cc|c} 2 & 1 & 3 \\ 1+2k & 2+k & 3+3k \end{array}\right] \begin{matrix} \leftarrow 2x_1 + x_2 = 3 \\ \leftarrow (1+2k)x_1 + (2+k)x_2 = 3(1+k) \leftarrow l_2^{(k)} \end{matrix}$$

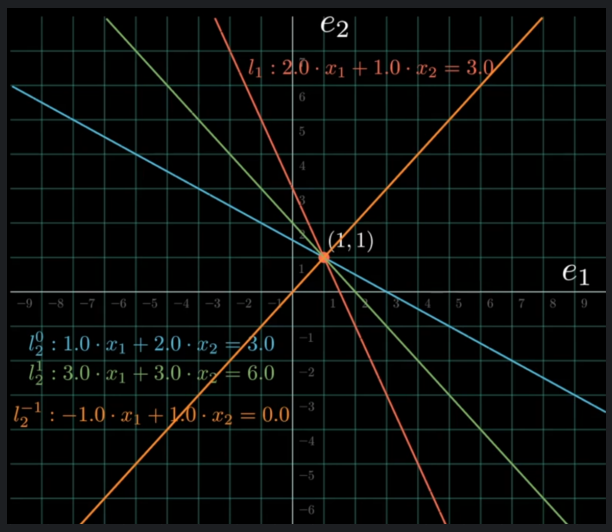
↳ plotting for diff values of k

$l_2^{(0)}$: $x_1 + 2x_2 = 3$

$l_2^{(1)}$: $3x_1 + 3x_2 = 6$

$l_2^{(-1)}$: $-x_1 + x_2 = 0$

$l_2^{(2)}$: $5x_1 + 4x_2 = 9$



Determinants

Any square matrix A_{nxn} is associated with a determinant function. $Det(A) \to \mathbb{R}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad det(A) = a_{11}a_{12} - a_{12}a_{21}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad det(A) = a_{11} \cdot det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12} \cdot det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13} \cdot det\left(\begin{bmatrix} a_{21} & a_{31} \\ a_{31} & a_{32} \end{bmatrix}\right)$$

$det(A_{2x2})$ = Area of 11^{gm} determined by the col^{m} vectors $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}, \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$.

$det(A_{3x3})$ = Area of 11^{ped} $(A_1), (A_2), (A_3)$.

Note:
- det A_{nxn}, where A is a diagonal matrix, is the product of the diagonal elements.
- det A_{nxn}, where A is an upper △ar matrix, is the product of the diagonal elements.

$$A = \begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix} \quad det(A) = d_1 \cdot d_2 \cdots d_n = \prod_{i=1}^{n} d_i$$

$$det(U) = \prod_{i=1}^{n} u_{ii}$$

- A_{nxn}, for any scalar c, $det(cA) = c^n det(A)$.
- If det(A) = 0, then A is called singular.
- If $det(A_{nxn}) \neq 0$, then rank(A) = n.

For a singular matrix A_{nxn}, the following statements are equivalent.
- A_{nxn} is singular.
- Rank(A) < n.
- det(A) = 0.
- A is NOT row equivalent to I_{nxn}.
- Ax=0 has non-trivial solution.
- Ax=b does NOT have a unique solution (has none or ∞ly many solutions).

For a non-singular matrix A_{nxn}, the following statements are equivalent.
- A_{nxn} is non-singular.
- Rank(A) = n.
- $det(A) \neq 0$.
- A is row equivalent to I_{nxn}.
- Ax=0 has only the trivial solution.
- Ax=b has a unique solution x for every given b.

- $det(I_{nxn}) = 1$.
- For any matrix A with 2 equal rows, det(A) = 0.
- $det(A^T) = det(A)$.
- det(AB) = det(A)·det(B).

Row swap

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad det(A) = a_{11}a_{22} - a_{12}a_{21}$$

$(R_1 \leftrightarrow R_2)$

$$A_{RS} = \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix} \qquad det(A_{RS}) = a_{21}a_{12} - a_{11}a_{22} = -det(A)$$

Row swapping once changes the sign of the determinant.

→ odd # of swaps ⇒ $-det(A)$
even # of swaps ⇒ $det(A)$

$det(A+B) \neq det(A) + det(B)$, there could be some exceptional cases, but not always.

Matrix inverse

For $a \in \mathbb{R}$, $a \neq 0$, we can find the multiplicative inverse $a^{-1} \in \mathbb{R}$ s.t. $a \cdot a^{-1} = a^{-1} \cdot a = 1$ (unique for a).

Ily, for a matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, we want B s.t. $AB = I$. Does such a B exist?

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

If $AB = I$, $a_{11}b_{12} = -a_{12}b_{22} \Rightarrow b_{12} = -\frac{a_{12}}{a_{11}} b_{22}$

$a_{21}b_{11} = -a_{22}b_{21} \Rightarrow b_{21} = -\frac{a_{21}}{a_{22}} b_{11}$

$$\Rightarrow b_{11} = \frac{a_{22}}{a_{11}a_{22} - a_{12}a_{21}} \; ; \; b_{22} = \frac{a_{11}}{a_{11}a_{22} - a_{12}a_{21}}$$ → $\det(A)$

$$b_{12} = \frac{-a_{12}}{a_{11}a_{22} - a_{12}a_{21}} \; ; \; b_{21} = \frac{-a_{21}}{a_{11}a_{22} - a_{12}a_{21}}$$

$$A^{-1} = B = \frac{1}{\det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

$A^{-1}A = AA^{-1} = I$

$\det(A) \neq 0 \Rightarrow A^{-1}$ exists.

$A_{m \times n}$, $m \neq n \Rightarrow A^{-1}$ is not defined.

Properties of inverses

If A^{-1} and B^{-1} exist, A and B are of the same size,

$(AB)^{-1} = B^{-1}A^{-1}$

Proof: Let $AB = C$.

$C^{-1} = (AB)^{-1}$

$C \cdot C^{-1} = I$

$AB\,C^{-1} = I$

$A^{-1}AB\,C^{-1} = A^{-1}I$

$I\,BC^{-1} = A^{-1}I = A^{-1}$

$B^{-1}BC^{-1} = B^{-1}A^{-1}$

$I\,C^{-1} = B^{-1}A^{-1}$

- Product of any # of invertible matrices is invertible, and inverse of product = product of inverses reversed

  $(ABCD\ldots)^{-1} = \ldots D^{-1}C^{-1}B^{-1}A^{-1}$

- $(A^{-1})^{-1} = A$
- $c \neq 0$, cA is also invertible if A is invertible. $(cA)^{-1} = c^{-1}A^{-1}$
- If A is invertible, $(A^T)^{-1} = (A^{-1})^T$

For $A_{n\times n} = 0$, $\det(A) \neq 0$, $A^{-1} = \dfrac{\text{Adj}(A)}{\det(A)}$ $\quad \text{adj}(A) = \text{adjoint of } A$

$$\text{Adj}(A) = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \qquad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\text{adj}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Computation of A^{-1} is quite tedious. So Gauss-Jordan elimination is used.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \qquad B = A^{-1} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

$$AB = I$$



we can find RREF$[A|I]$ to get A^{-1}.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 3 & 1 & 2 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \leftarrow R_2 - 3R_1, \quad R_3 \leftarrow R_3 - 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -1 & -5 & -2 & 0 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - \dfrac{R_2}{5}$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & 0 & -\frac{18}{5} & -\frac{2}{5} & -\frac{1}{5} & 1 \end{array}\right]$$

$R_3 \leftarrow \dfrac{R_3}{-\frac{18}{5}}$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & 0 & 1 & \frac{7}{18} & \frac{1}{18} & -\frac{5}{18} \end{array}\right]$$

$R_2 \leftarrow R_2 + 7R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -5 & 0 & -\frac{5}{18} & \frac{25}{18} & -\frac{35}{18} \\ 0 & 0 & 1 & \frac{7}{18} & \frac{1}{18} & -\frac{5}{18} \end{array}\right]$$

$R_1 \leftarrow R_1 - 3R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -\frac{3}{18} & -\frac{3}{18} & \frac{15}{18} \\ 0 & -5 & 0 & -\frac{5}{18} & \frac{25}{18} & -\frac{35}{18} \\ 0 & 0 & 1 & \frac{7}{18} & \frac{1}{18} & -\frac{5}{18} \end{array}\right]$$

$R_1 \leftarrow R_1 + \dfrac{2}{5} R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -5/18 & 7/18 & 1/18 \\ 0 & -5 & 0 & -5/18 & 25/18 & -35/18 \\ 0 & 0 & 1 & 7/18 & 1/18 & -5/18 \end{array}\right]$$

$R_2 \leftarrow \dfrac{R_2}{-5}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -5/18 & 7/18 & 1/18 \\ 0 & 1 & 0 & 1/18 & -5/18 & 7/18 \\ 0 & 0 & 1 & 7/18 & 1/18 & -5/18 \end{array}\right]$$

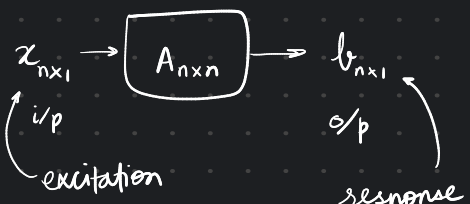
I $\qquad$ A^{-1}

$$A^{-1}A = \begin{bmatrix} -5/18 & 7/18 & 1/18 \\ 1/18 & -5/18 & 7/18 \\ 7/18 & 1/18 & -5/18 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} = I$$

Interpreting the inverse

Consider $Ax=b$, a system of n linear eq^{n}s in n unknowns.

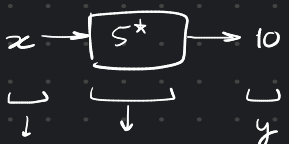
↳ $A_{n\times n}\, x_{n\times 1} = b_{n\times 1}$



consider A as a system that takes $x_{n\times 1}$ and gives $b_{n\times 1}$.

When we say solve $Ax=b$,
we have a system A, we want an output (b),
we want to know what input gives the req'd output b.
↳ x

A simple example: $y = 5*x = 10$



$ax = y$

To find what x makes 5*x the output 10, we take inverse of the system and multiply it by 10.

$5x = 10$

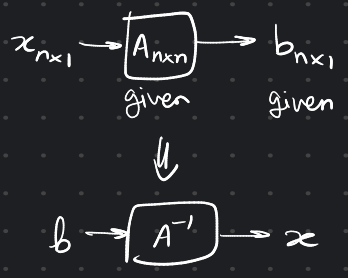
$x = \frac{1}{5} * 10$

$\frac{1}{5}$ ↳ inverse; 10 ↳ output

Given y, we can always find $x = \frac{1}{5} y$

$x = a^{-1} y$

Ily, for $Ax=b$, a set of eq^{n}s in n unknowns.



$x_{n\times 1} = A^{-1}_{n\times n}\, b_{n\times 1}$

Reformulation of $Ax=b$: Find the output of the system A^{-1} corresponding to the input b.

$a_{11}x_1 + a_{12}x_2 = b_1$
$a_{21}x_1 + a_{22}x_2 = b_2$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \boxed{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} \rightarrow \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

$a_{21}x_1 + \frac{a_{21}a_{12}}{a_{11}} x_2 = \frac{a_{21}}{a_{11}} b_1$

$\left(a_{22} - \frac{a_{21}a_{12}}{a_{11}}\right) x_2 = b_2 - \frac{a_{21}}{a_{11}} b_1$

$(a_{11}a_{22} - a_{21}a_{12})\, x_2 = a_{11}b_2 - a_{21}b_1$

$x_2 = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{21}a_{12}}$

Ily, $x_1 = \frac{a_{22}b_1 - a_{12}b_2}{a_{11}a_{22} - a_{21}a_{12}}$

$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22}b_1 - a_{12}b_2 \\ a_{11}b_2 - a_{21}b_1 \end{bmatrix}$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

$x = A^{-1} b$

$b \rightarrow \boxed{A^{-1}} \rightarrow x$

Cramer's rule

Suppose $Ax = b$. $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$.

$$a_{11}x_1 + a_{12}x_2 = b_1 \quad \text{—①}$$
$$a_{21}x_1 + a_{22}x_2 = b_2 \quad \text{—②}$$
$$① \times \frac{a_{21}}{a_{11}} \rightarrow a_{21}x_1 + \frac{a_{21}a_{12}}{a_{11}}x_2 = \frac{a_{21}b_1}{a_{11}} \quad \text{—③}$$

② − ③

$$(a_{22}a_{11} - a_{21}a_{12})x_2 = a_{11}b_2 - a_{21}b_1$$
$$x_2 = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{21}a_{12}} \leftarrow \det(A)$$
$$x_1 = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{21}a_{12}} \leftarrow \det(A)$$

$$x_1 = \frac{b_1a_{22} - b_2a_{12}}{\det(A)} \qquad x_2 = \frac{a_{11}b_2 - a_{21}b_1}{\det(A)}$$

for x_1, replace col^m 1 with RHS.

$$x_1 = \frac{\det\begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix}}{\det(A)} \qquad x_2 = \frac{\det\begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}}{\det(A)}$$

for x_2, replace col^m 2 with RHS.

$a_{11}x_1 + a_{12}x_2 = b_1$
$a_{21}x_1 + a_{22}x_2 = b_2$

x_1 is multiplied by 1st col^m $\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}$ $\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$ $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ x_2 is multiplied by 2nd col^m.

For an $n \times n$ matrix $A^{n \times n}$,

$$A^{n \times n} x^{n \times 1} = b^{n \times 1}$$

Still x_i is multiplied by i^{th} col^m of A.

Cramer's rule: $A^{n \times n} x^{n \times 1} = b^{n \times 1}$ If $\det(A) \neq 0$, $\text{rank}(A) = n$

$$x_i = \frac{\det([A_1\ A_2 \cdots \underset{i^{th}\ \text{col}^m}{b} \cdots A_n])}{\det([A_1\ A_2 \cdots A_i \cdots A_n])} \qquad \text{for } i = 1 \ldots n$$

Again, this is computationally expensive for large matrices, so Gaussian elimination is preferred.

$\det(A^{2 \times 2})$: Area of $\parallel^{gm}$ whose sides are vectors along col^m vectors of A.

Cramer's rule: $\dfrac{\text{Area of some } \parallel^{gm}}{\text{Area of some other } \parallel^{gm}}$

Suppose we have $Ax = 0$. $a_{11}x_1 + a_{12}x_2 = 0$, $a_{21}x_1 + a_{22}x_2 = 0$

$$x_1 = \frac{\det\begin{bmatrix} 0 & a_{12} \\ 0 & a_{22} \end{bmatrix}}{\det(A)} = 0 \qquad x_2 = \frac{\det\begin{bmatrix} a_{11} & 0 \\ a_{21} & 0 \end{bmatrix}}{\det(A)} = 0 \qquad \det A \neq 0$$

so when $\det A \neq 0$, $(x_1, x_2) = (0, 0)$ is the only soln for $Ax = 0$.

To apply Cramer's rule, A must be non-singular.

e.g. $2x_1 + x_2 = 3$, $2x_1 + x_2 = 4$ $\quad A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$ $\quad \det A = 0$ $\quad$ Cramer's rule won't work.

Ex. 2. $\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ 2x_1 + x_2 - x_3 &= 2 \\ x_1 - x_2 + x_3 &= 1 \end{aligned} \longrightarrow A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$\det(A) = 1(1-1) - 1(2+1) + 1(-2-1) = -6$

$x_1 = \dfrac{\det \begin{bmatrix} 3 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}}{\det(A)} = \dfrac{-6}{-6} = 1 \qquad x_2 = \dfrac{\det \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}}{\det(A)} = \dfrac{-6}{-6} = 1$

Ily, $x_3 = 1$.